



Choosing RAM modules

Here are key factors when selecting kits:

1. **Compatibility:** Ensure your motherboard's chipset supports DDR5 and check the QVL (qualified vendor list) for tested modules. Intel's Z890 and AMD's X670/B650 boards support DDR5.
2. **Brand and build quality:** Partner brands like Corsair Vengeance/Dominator, Kingston FURY Beast/Renegade, G.Skill Trident Z5/Neo, Crucial Pro, and ADATA XPG Lancer have strong reputations. Many offer integrated heat spreaders for better thermals and optional RGB lighting. Don't pay extra solely for aesthetics; performance and warranty coverage are more important.
3. **Speed vs latency:** Evaluate kits by both frequency and timings. A DDR5-6000 CL30 kit often outperforms a DDR5-7000 CL40 kit due to lower latency. High frequencies above 7200 MT/s may require manual tuning and deliver diminishing returns because of gear modes.
4. **Overclocking headroom:** Enthusiast kits with binned ICs (e.g., Corsair Dominator Titanium, G.Skill Trident Z5 RGB) often overclock beyond their rated speeds. If you plan manual overclocking, choose modules built with high-quality chips and robust PCB layouts.
5. **Aesthetics and features:** RGB lighting, black or white heat spreaders, and low-profile designs (helpful under large air coolers) are personal preferences. Brands like Kingston FURY

Renegade and XPG Lancer integrate diffused RGB lighting controlled via motherboard software.

Example Configurations

- **Mainstream Gaming PC:** Two 16 GB modules of DDR5-6000 CL30, such as Corsair Vengeance RGB or Kingston FURY Beast. Install them in slots A2 and B2 to enable dual-channel.
- **Creative Workstation:** Four 16 GB modules (total 64 GB) DDR5-6000 CL32, such as G.Skill Trident Z5 RGB. Choose kits with EXPO profiles if using Ryzen 9000 for easiest setup.
- **Compact ITX Build:** Two 24 GB modules (48 GB total) DDR5-6400 CL32 like ADATA XPG Lancer Blade, designed with low-profile heat spreaders to clear the cooler. Ensure your ITX board's BIOS is updated for 24 GB module support.

Conclusion

Memory has evolved from a commodity component to a performance differentiator. DDR5 brings fundamental architectural changes – independent channels, more bank groups, on-die ECC and integrated power management – that unlock the full potential of modern multi-core CPUs. Decoding specifications and understanding the interplay between speed and latency helps you choose a kit that suits your workload. Enabling XMP or EXPO is essential to realise the speeds printed on the box, and the right capacity ensures a smooth computing experience whether you're gaming, creating or crunching numbers. **d**